



Institute of Health
Department of Biomedical Sciences
Biochemistry for Pre-clerkship –I
2021/2022 Entry

Objective of the session

- At the end of this session students will be able:-
 - ✓ Describe Composition of bone
 - ✓ Discuss Vitamin D, Phosphorus and Calcium metabolism.
 - ✓ Explain Manifestation of hyper and hypocalcaemia osteoporosis, Osteomalacia, Rickets and Scurvy.
 - ✓ Elaborate Collagen biosynthesis and its disorders

Bone Metabolism

➤ Cells :

- ✓ Osteoblasts- Bone forming cells.
- ✓ Osteocytes - maintain bone
 - Constitute 90%-95% of cells in mature skeleton.
- ✓ Osteoclasts- These cells helps in bone resorption.

➤ Matrix:

- ✓ Organic matrix= mainly type I collagen fibers
- ✓ Inorganic matrix= calcium, magnesium, phosphate, carbonate, chloride, fluoride and citrate.

Cont...

❖ Inorganic content

- ✓ It gives rigidity to the bone
- ✓ Its deficiency causes soft bone(rickets/osteomalacia)

❖ Organic content

- ✓ It gives the elasticity to the bone.
- ✓ Lack of them cause brittle bone (Osteogenesis Imperfect/Osteoporosis)

❖ What is the composition of bone?

- ✓ 60% inorganic

- Calcium hydroxyapatite

- ✓ 40% organic(non mineralized matrix is known as osteoid)

- Type 1 collagen (tensile strength)

- Proteoglycans (compressive strength)

- Osteocalcin/Osteonectin(bone forming protein)

- Growth factors/Cytokines/Osteoid

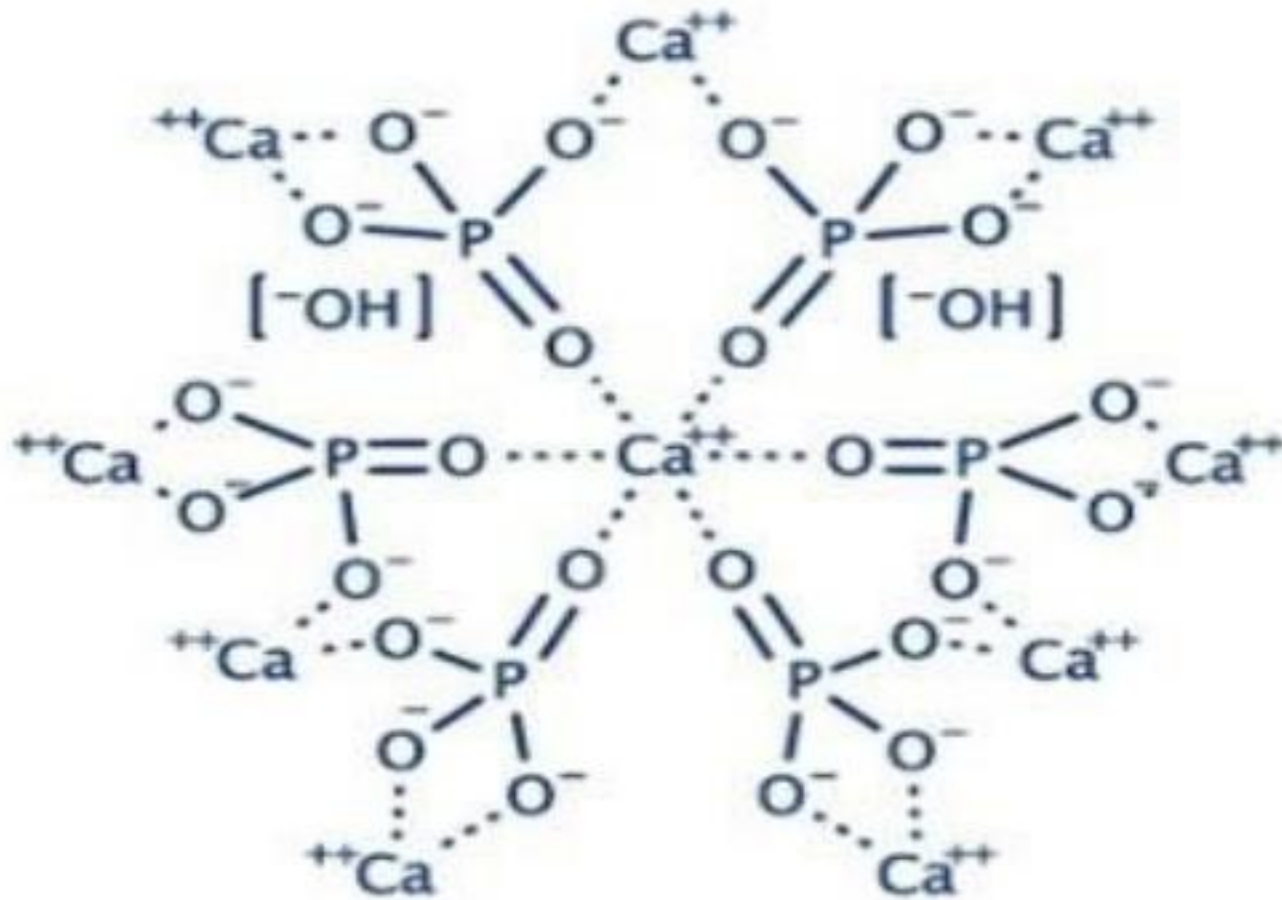


Fig1. Structure of calcium in the skeleton-calcium phosphate (hydroxyapatite)

Calcium metabolism

- What is the recommended daily intake?
 - ✓ 1000mg(for avg adult)
- Total body calcium?
 - ✓ 1100gm(skeleton-99% & plasma-1%)
- What is the plasma concentration?
 - ✓ 2.2-2.6mmol/L
- How is calcium excreted?
 - ✓ Kidneys - 2.5-10mmol/24 hrs
- *Hypercalcium and Hypocalcium???*
- How are calcium levels regulated?
 - ✓ PTH, calcitonin and vitamin D (+others)

Cont...

➤ Function of Ca:

- ✓ coagulopathy,
- ✓ nerve function,
- ✓ contraction of muscle.

➤ **Ca** in bone occurs in two forms :

- ✓ readily exchangeable (500mmol/day) and
- ✓ slowly exchangeable (7.5mmol/day) which is **more stable form and concerned with bone remodeling by constant resorption & deposition**

Calcium Homeostasis

- A large amount of **Ca** is filtered in kidneys & 98-99% is reabsorbed (60% in proximal tubule and rest in ascending limb of loop of henle and regulated in turn by PTH)
- **Ca** absorbed from the gastrointestinal (GI) tract is actively transported out of the intestine by system in brush border involving **Ca** dependent ATPase and regulated by 1,25- DHCC.
- Some absorption also occurs by passive diffusion.
- When **Ca** intake is high - 1,25-DHCC level falls and increase with deficient **Ca** intake.

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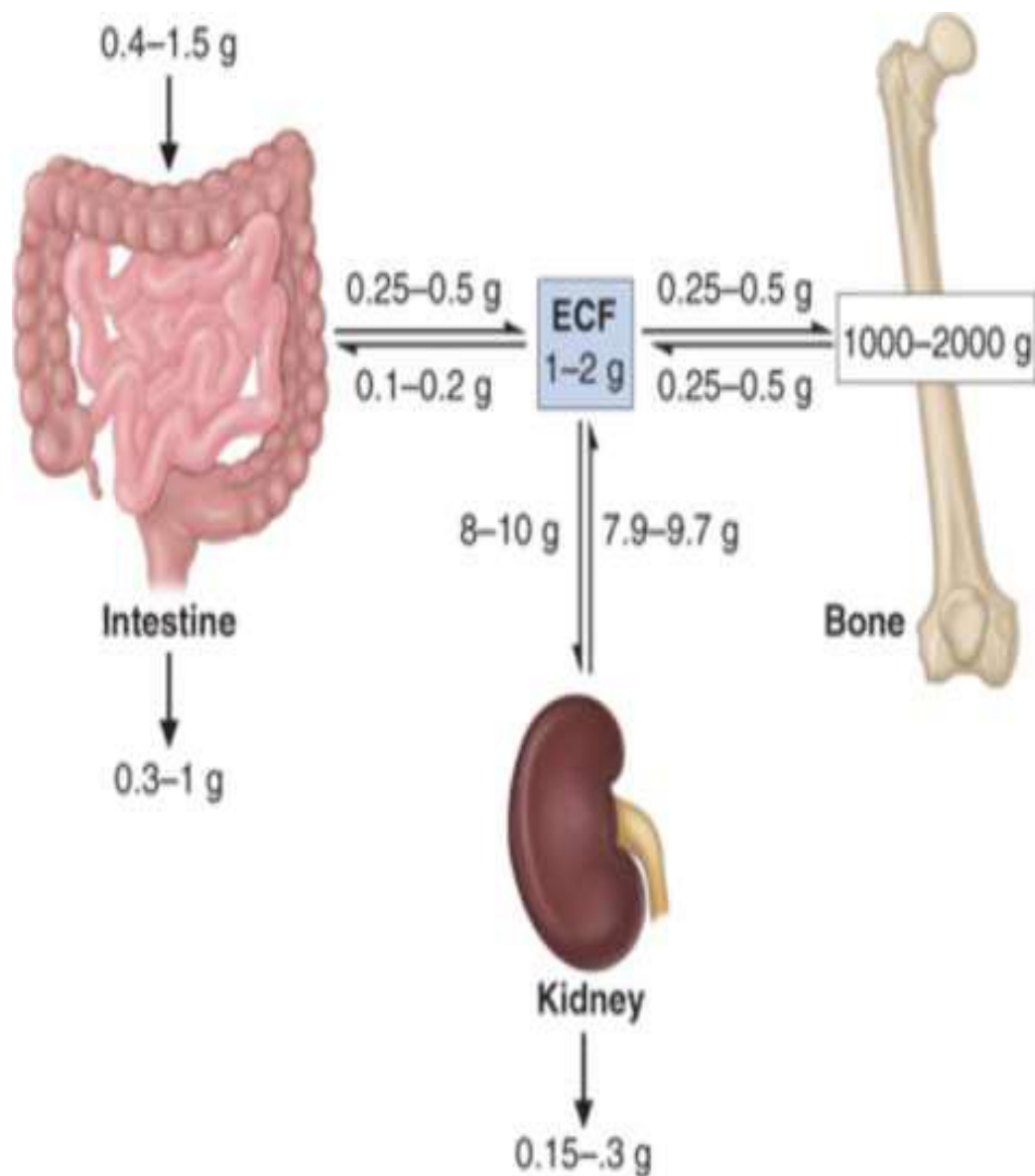
❖ Determined by several factors:

➤ Entries

- ✓ Intestinal absorption (variable between 0.4 and 1.5 g per day)
- ✓ Renal reabsorption (variable between 7.9 and 9.7 g per day, equal to about 98% of the Ca^{2+} filtered)

➤ Outputs

- ✓ Intestinal excretion (variable between 0.3 and 1 g per day)
- ✓ Renal excretion (ranging from 0.15 to 0.3 g per day)
- ✓ Exchange between ECF BONE (0.25 - 0.5 g / day)



Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. Loscalzo: Harrison's Principles of Internal Medicine, 19th Edition.
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Calcium Homeostasis

❖ Hormonal serum Ca^{2+} control obtained by:-

- ✓ Parathyroid hormone
- ✓ Calcitonin
- ✓ 1,25-dihydroxycholecalciferol (1,25-DHCC)

❖ Their target organs are:-

- the intestine
- Kidney and
- bone

Cont...

I. Parathyroid hormone

- ✓ Hormone produced by parathyroid
- ✓ PTH causes the increase of plasma calcium concentration
- ✓ Parathyroid cells exhibit a receptor able to bind Ca^{2+} (GPCR - G protein coupled receptor)
- ✓ The GPCR has an extensive extracellular domain that binds Ca^{2+} ,
- ✓ the ion binding activates the receptor that has inhibitory effect on the secretion of PTH;
- ✓ as Ca^{2+} bound to the receptor decreases, the inhibition falls causing a ready secretory response of PTH

Cont...

- ✓ **PTH** after interact with the type I receptor (PTH1R); **carried out different activities** to increases plasma Ca^{2+} at different tissues:-

At the level of **the bone**

- it determines stimulation of osteoclasts with resorption of the bone matrix with release of Ca^{2+}

At the **Kidney** perform two things

- it increases the tubular reabsorption of Ca^{2+} in distal convoluted tubule (DCT).
- It increases the activity of the 1-hydroxylase resulting in increased synthesis of **1,25-DHCC (vit.D)** in the kidney

Cont...

- ✓ Its secretion is stimulated by the reduction of calcium concentration in the extracellular fluid and is inhibited by its increase and by 1,25-DHCC
- ✓ Circulating hormone is cleared by the liver and the kidney

Cont...

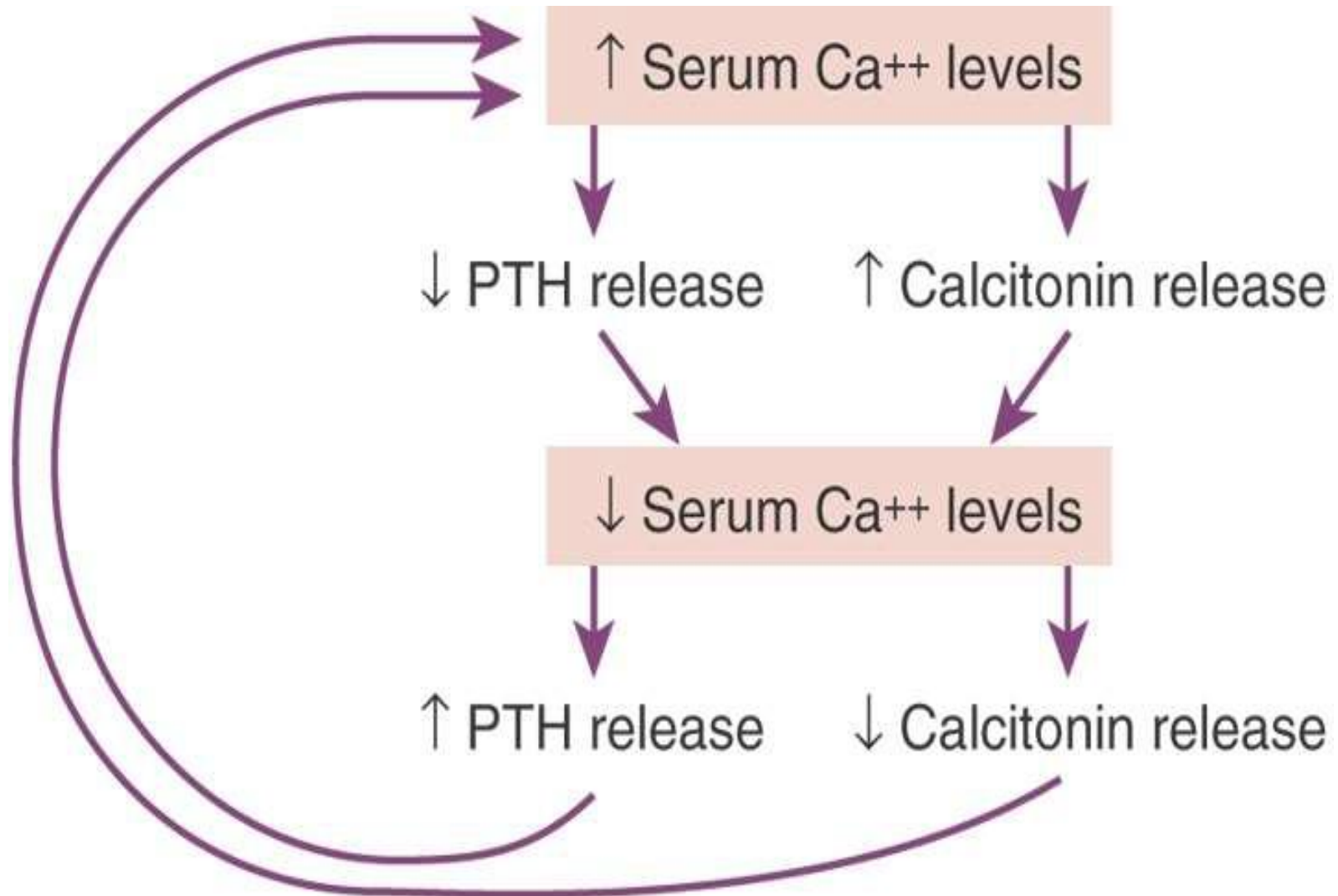
II. Calcitonin

- ✓ Produced by the thyroid gland cells in response to high Ca^{++} levels in the blood:
- ✓ Decreases the concentration of Ca^{++} in the blood
- ✓ At bone
 - ✓ stimulate osteoblast and inhibiting osteoclast activity
- ✓ In Kidney
 - ✓ Enhance excretion of both calcium and phosphorus
- ✓ *Counteract the calcium-producing effects of PTH*

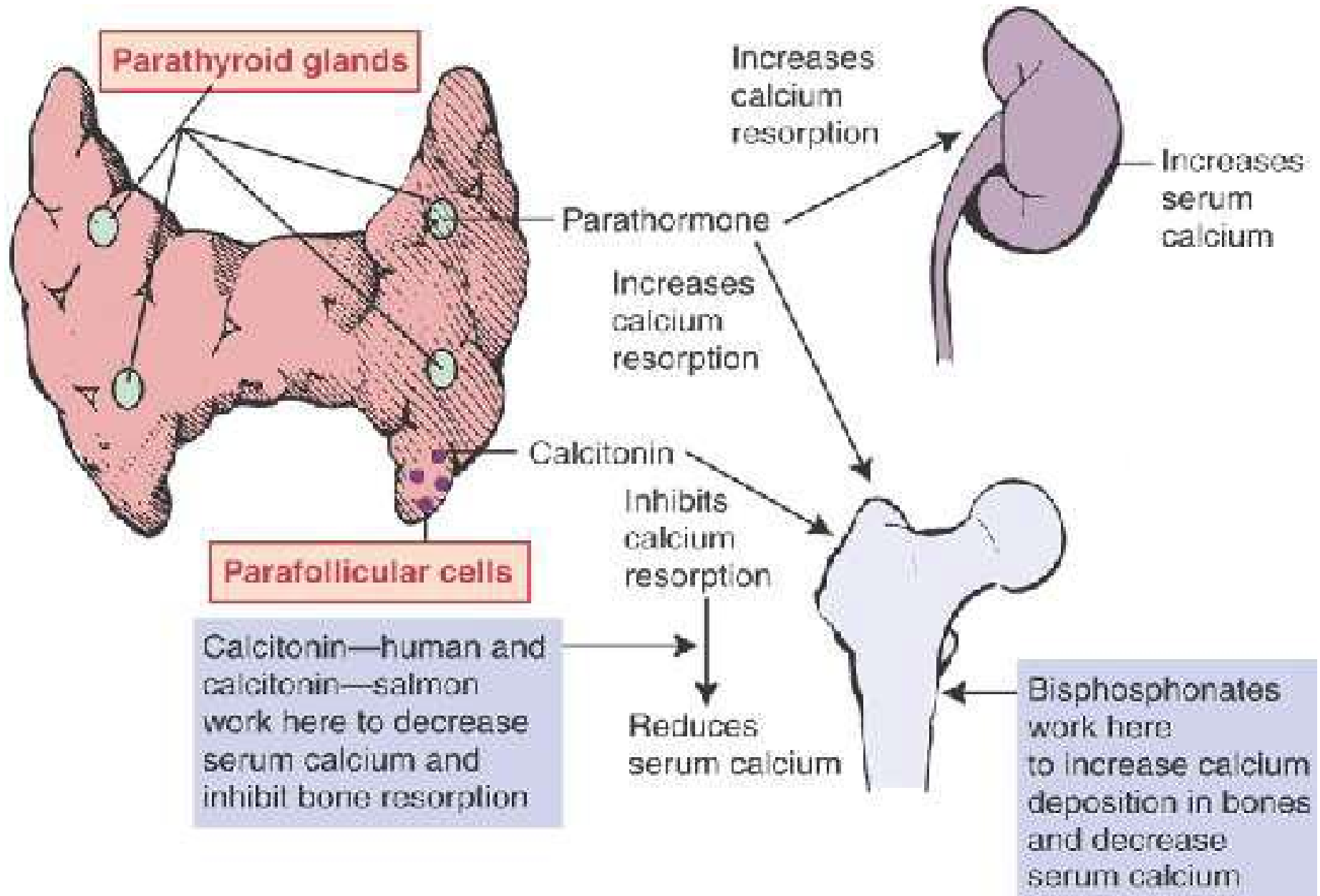
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- **Fibroblast growth factor 23 (FGF23)** helps regulate **plasma phosphate concentration**.
 - ✓ High **phosphate and 1,25-dihydroxyvitamin D** levels are FGF23 stimulators
 - ✓ FGF23 decreases the production of cotransporters in the kidney so that increases phosphate renal excretion
 - ✓ it (FGF23) downregulates the production of 1,25-dihydroxyvitamin D, subsequently causes decreases gut absorption
 - ✓ **Eventually** lower serum phosphorous

Regulation of PTH and Calcitonin Secretion



PTH and Calcitonin



Parathyroid Hormone disorders

➤ Hyperthyroidism

➤ Hypoparathyroidism

Hyper parathyroidism

- Increase PTH
- Blood Ca^{++} increased & phosphate decreased.
- ❖ Sign and Symptoms(manifestations)
 - ✓ Hypercalcaemia, kidney stones (nephrocalcinosis)
 - ✓ Osteoporosis, osteomalacia, & arthritis.
 - ✓ Lethargy, fatigue, depression, memory loss, psychosis, ataxia, delirium, & coma.

Hypo parathyroidism

➤ Hypo parathyroidism = reduction of PTH

✓ Causes Hypocalcaemia

❖ Signs and symptoms(manifestations):

✓ Muscle cramps, muscle spasms called tetany (can lead to spasms of the larynx, causing breathing difficulties).

✓ Pain in the face, legs, feet & abdomen.

✓ Dry hair & skin, brittle nails, cataracts.

✓ Weakened tooth enamel (in children).

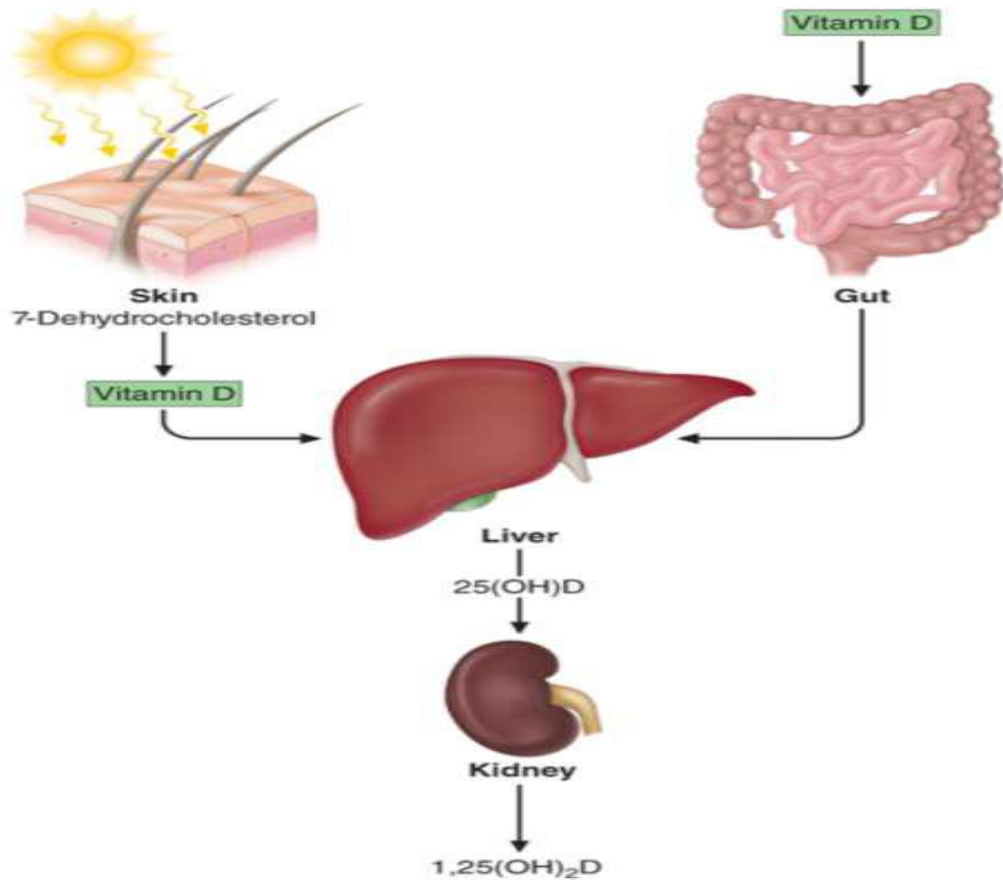
✓ Convulsions (seizures).

III. 1,25-DIHYDROXYCHOLECALCIFEROL (Vitamin D)

- ✓ It is considered as a hormone
- ✓ Also it can be synthesized in the skin and gained from some foods
- ✓ It increases calcium absorption and deposition in the bone
- ✓ It requires activation by *double hydroxylation*
 - ❖ *The first, in position 25, is by the liver and is not subject to any control*
 - ❖ *the second, in position 1, is by alpha 1-hydroxylase in the kidney and is controlled by PTH*

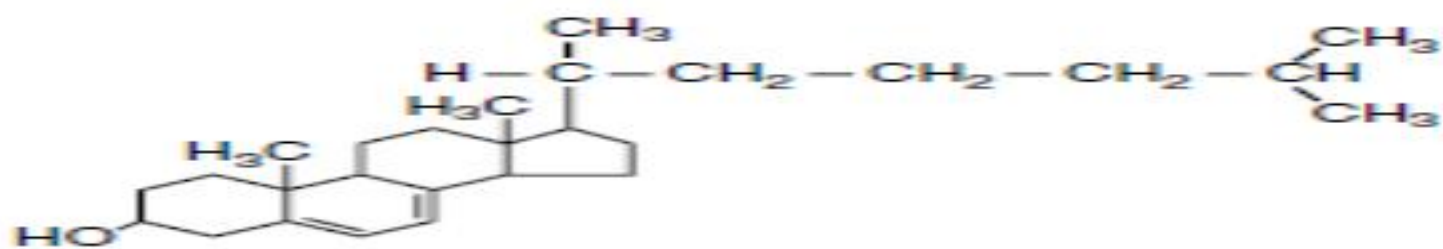
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- ✓ The activity of the 1-hydroxylase is increased by PTH and by the reduction of Ca^{2+} concentration in ECF while it is inhibited by the product of the reaction (1,25-DHCC)
- ✓ It is present in the circulation bound to a globulin synthesized by the liver and to albumin
- ✓ It is **inactivated** by hydroxylation in position 24 and eliminated in the gall



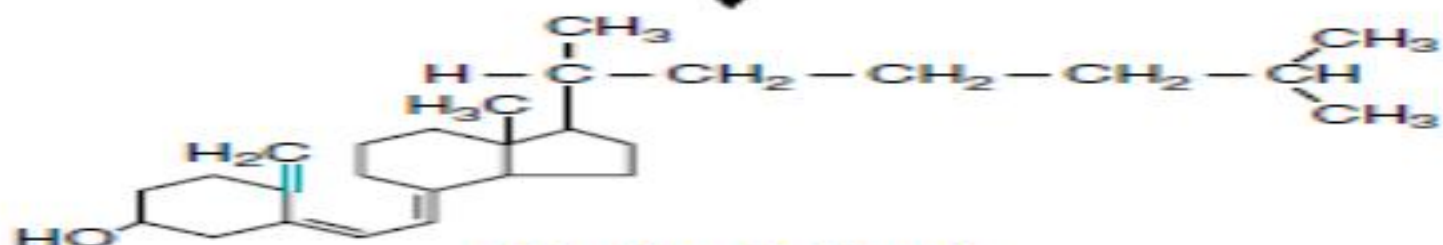
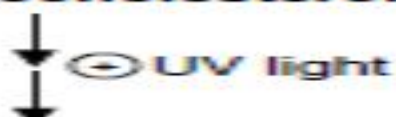
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Vitamin D synthesis and activation



7-Dehydrocholesterol

Skin

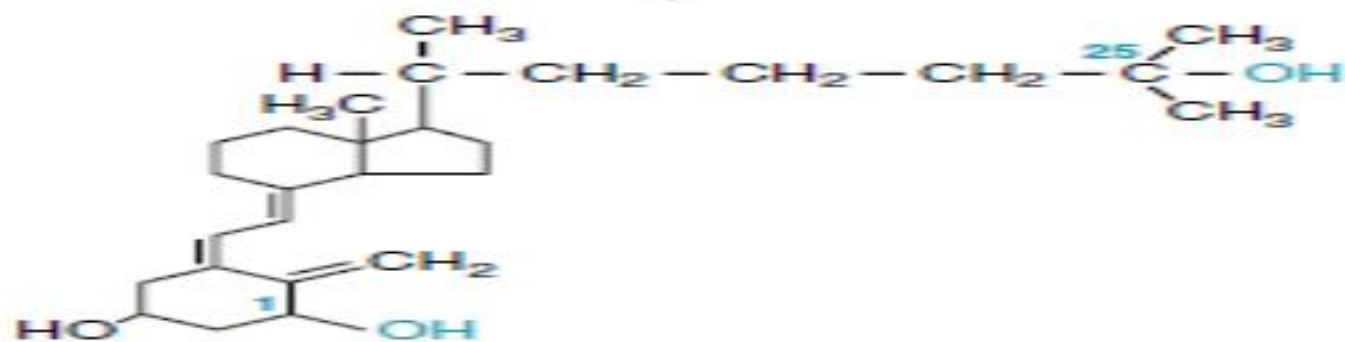


Cholecalciferol

Liver

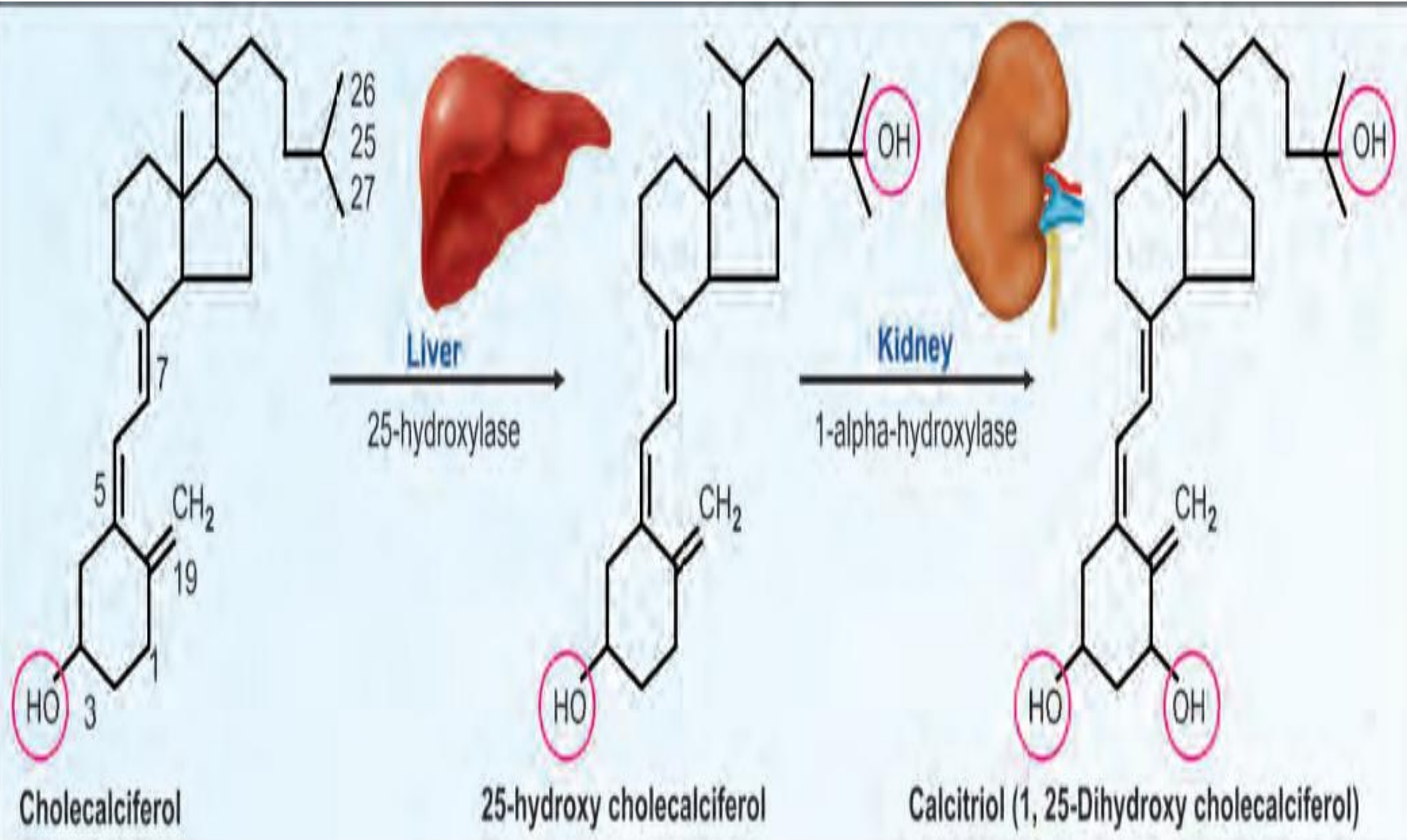
25-Hydroxycholecalciferol

Kidney



1,25-Dihydroxycholecalciferol
(1,25-(OH)₂D₃)

Cont...



EFFECTS OF VITAMIN D

- ✓ It acts by binding its specific receptor (VDR) belonging to the nuclear receptor superfamily
- ✓ VDR activation is followed by nuclear translocation and induction of the expression of target genes

Target organs:

- Intestinal
- Bone
- Kidney

Cont...

- **Intestinal** : increased expression of membrane **Ca²⁺ channels** → promotes the absorption of calcium and phosphorus from the intestine
- **Bone**: **Mineralization** of the bone is increased by increasing the activity of osteoblasts → increases bone mineral density
 - ✓ Active vitamin D has action on all three types of bone cells—osteoblasts, osteoclasts and osteocytes
- ✓ **Kidney**: increases the reabsorption of **calcium and phosphorus by renal tubules**, therefore both minerals are conserved

Deficiency of Vitamin D

- The deficiency diseases are **rickets** in children and **osteomalacia** in adults. Hence vitamin D is known as antirachitic vitamin.
- The optimal concentration of 25-hydroxy D3 is **> 30ng/mL**, where as **20-29ng/mL** is considered **insufficient** and **10-19ng/mL** is **deficient**.
- A level below **10ng/mL** indicates **severe deficiency**
- **Rickets**:-the bones of children are **undermineralized** as a result of poor absorption of calcium.
- **Osteomalacia**(softens bones):- in adults results from **demineralization** of bone, more common in women & often happens during pregnancy.
- **Osteoporosis** is the **weakening of the bone**

Osteomalacia	Osteoporosis	Differences
Demineralization of the bones	Reduction of the bone mineral density	Definition
Adults	≥ 65 years women	Presence
Weakness of the muscles and fragile bones	Curving of the back and risk for fractures	signs and symptoms
Deficiency in Vitamin D	Deficiency in calcium and phosphorous, drugs, inherited, endocrine disorders, alcohol drinking.	Causes
Injections of Vitamin D	Healthy lifestyle, calcium supplements and naturally calcium-diet	Treatment

Cont...

➤ Causes for Vitamin D Deficiency

❖ Deficiency of vitamin D can occur in people

1. **Not exposed to sunlight properly**, e.g. inhabitants of northern latitudes, in winter months, in people who are bedridden for long periods.
2. **Nutritional deficiency of calcium or phosphate** may also produce similar clinical picture.
3. **Malabsorption of vitamin** (obstructive jaundice and steatorrhea)
 - High phytate content in diet may also reduce the absorption of vitamin.
4. **Abnormality of vitamin D activation**
 - Liver and renal diseases may retard the hydroxylation reactions.

Clinical Features of Rickets

- I. Rickets is seen in children. There is insufficient mineralization of bone. Bones become soft and pliable. The bone growth is markedly affected.
- II. The classical features of rickets are **bone deformities**. Weight bearing bones are bent. Continued action of muscles also cause bone malformations.
- III. The clinical manifestations include bow legs, knockknee, rickety rosary, bossing of frontal bones, and pigeon chest.
- IV. An enlargement of the epiphysis at the lower end of ribs and costochondral junction leads to beading of ribs or **rickety rosary**.
- V. **Harrison's sulcus** is a transverse depression passing outwards from the costal cartilage to axilla. This is due to the indentation of lower ribs at the site of the attachment of diaphragm.

Vitamin D deficiency diseases



Genu varum

Genu Valgum



Bone deformity in
rickets

Clinical Features of Osteomalacia

- I. The term is derived from Greek "osteon" = bone; and "malakia" = softness. The bones are softened due to insufficient mineralization and increased osteoporosis. Patients are more prone to get fractures.
- II. The abnormalities in biochemical parameters are a slightly lower serum calcium, and a low serum phosphate.

BONE PROTEINS

BONE PROTEINS

- ✓ The extracellular components of bone consist of a solid mineral phase in close association with an organic matrix, of which 90-95% is **type I collagen**.
- ✓ The noncollagenous portion of the organic matrix is heterogeneous and contains serum proteins (as albumin) and many locally produced proteins, (cell attachment/signaling proteins such as thrombospondin, osteopontin, and fibronectin; calcium-binding proteins such as matrix gla protein and osteocalcin; and proteoglycans such as biglycan and decorin).

Cont...

- **Collagen:** Most abundant protein in human body (25% of dry weight of adult)
 - ✓ Found as components of skin, connective tissue, blood vessel walls, & sclera & cornea of eye
 - ✓ Typical Collagen is long, rigid structure in w/c three polypeptides ("α-chains") are wound around one another in a rope-like **triple helix**

Cont...

- Collagen serves structural functions in d/t organs w/c dictates its type & organization:-

In ECM or vitreous humor of eye:

- ✓ Collagen may be dispersed as a gel that gives structural support

In cornea of eye:

- ✓ Collagen is stacked so as to transmit light with a minimum of scattering

In tendons:

- ✓ Collagen may be bundled in tight, parallel fibers that provide great strength

In bone:

- ✓ Collagen occurs as fibers arranged at angle to each other so as to resist mechanical shear from any direction

Cont...

Types of Collagen

- ✓ There are more than 25 collagen types, & additional proteins that have collagen-like domains
- ✓ Collagens can be organized into three groups, based on their location & functions in body
 - **Group-I: Fibril-forming collagens**
 - **Group-II: Network-forming collagens**
 - **Group III: Fibril-associated collagens**

Cont...

➤ Group-I: Fibril-forming collagens

- » Have rope-like triple helix structure
- » Collagen type I, II, & III
- » **Type I collagen fibers**
 - Are found in supporting elements of high tensile strength (Skin, bone, tendon, blood vessels & cornea)
- » **Type II collagen fibers**
 - are restricted to cartilaginous structure (Cartilage, intervertebral disk, vitreous body)
- » **Type III collagen fibers**
 - are prevalent in more distensible tissues (Blood vessels, fetal skin)

Cont...

Group-II: Network-forming collagens:

» Form 3-D mesh, rather than distinct fibrils

» Collagen type IV & VII

- Type IV
 - collagen assembles into a sheet or meshwork that constitutes a major part of glomerular basement membranes of Kidney
- Type VII
 - collagen assembles into a sheet or meshwork beneath stratified squamous epithelia

Cont...

- Group III: Fibril-associated collagens:

- » Bind to surface of collagen fibrils, linking these fibrils to one another & to other components in ECM

- » Collagen type IX & XII

TYPE	TISSUE DISTRIBUTION
	Fibril-forming
I	Skin, bone, tendon, blood vessels, cornea
II	Cartilage, intervertebral disk, vitreous body
III	Blood vessels, fetal skin
	Network-forming
IV	Basement membrane
VII	Beneath stratified squamous epithelia
	Fibril-associated
IX	Cartilage
XII	Tendon, ligaments, some other tissues

Structure of Collagen

- Its triple-helical structure that places many of its amino acid side chains on surface of helix
- Collagen is rich in **Proline(Pro)** & **Glycine(Gly)** amino acids
 - ✓ Pro facilitates formation of helical conformation of each α -chain b/c its ring structure causes "kinks" in peptide chain
 - ✓ Gly founds in every third position of polypeptide chain
 - ✓ Gly fits into restricted spaces where three chains of helix come together
 - ✓ Presence of Pro & Gly dictates that helical conformation of α -chain **cannot** be **α -helix**

Structure of Collagen

Repeating sequence of amino acids in triple helix **Gly-X-Y-**

Where:

X= Frequently Proline

Y= Often Hydroxyproline or sometimes, Hydroxylsine

❖ Hydroxyproline (Hyp):

- ✓ Results from hydroxylation of some **Proline** residue during post translational modification polypeptide chains
- ✓ Important in stabilizing triple-helical structure of collagen b/c it maximizes interchain hydrogen bond formation

❖ Hydroxylsine (Hyl)

- ✓ Results from hydroxylation of some **Lysine** residue during post-translational modification polypeptide chains

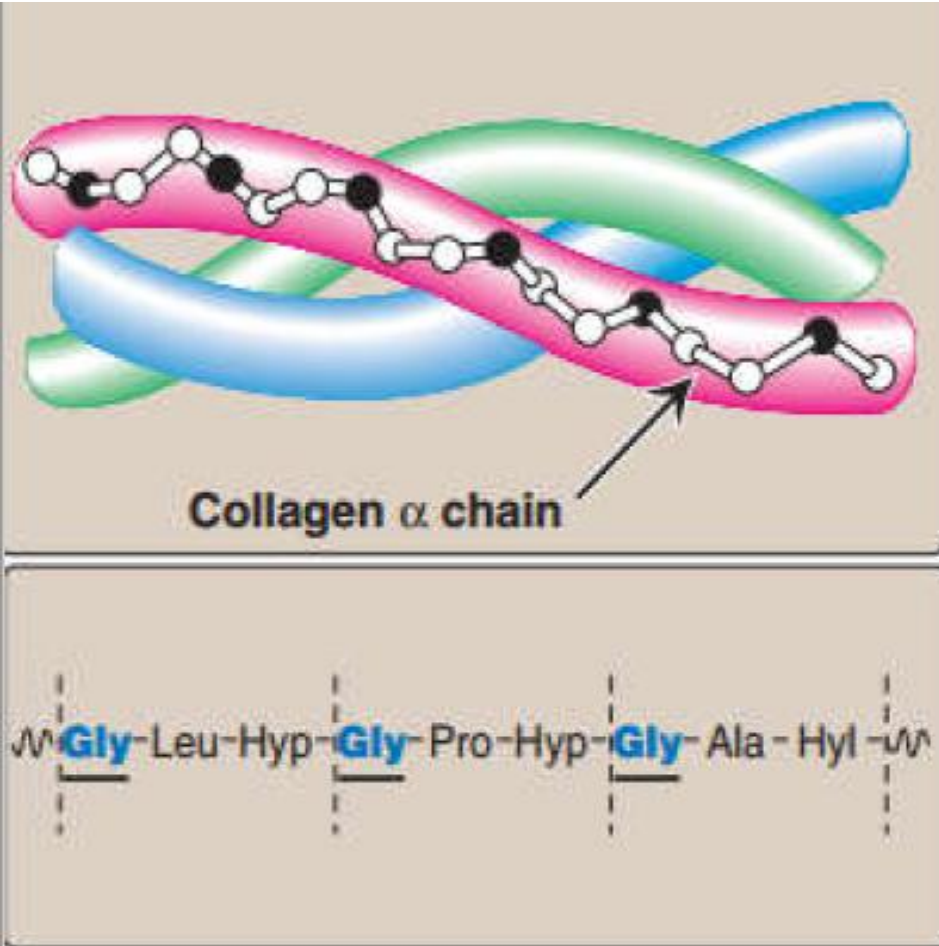


Fig1. Structure of Collagen

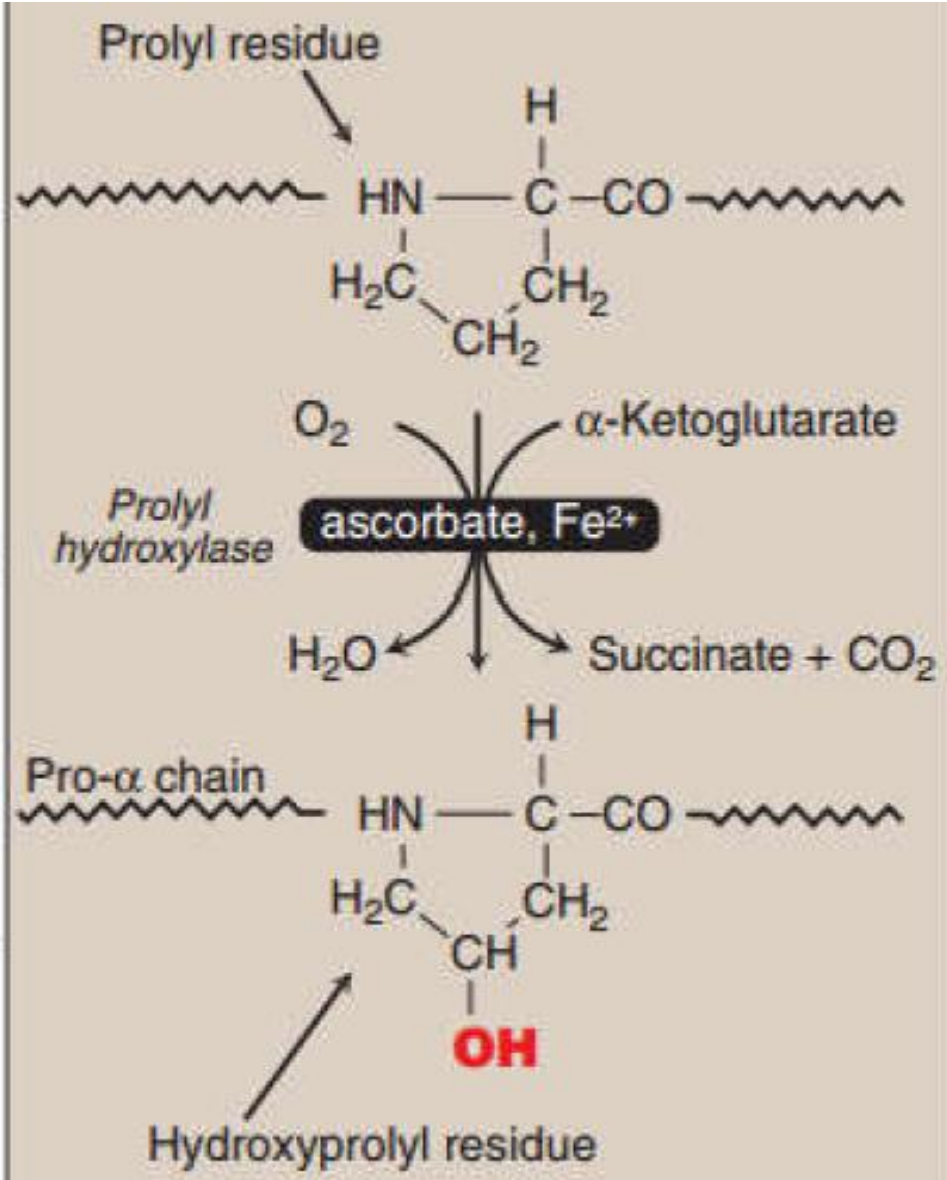


Fig2. hydroxylation reaction of **Proline** residue during post-translational modification

Cont...

- ✓ Polypeptide precursors of collagen:
 - Formed in fibroblasts of connective tissues or **osteoblasts of bone** & chondroblasts of cartilage
 - Secreted into extracellular matrix (ECM)
- ✓ After enzymatic modification, mature collagen monomers aggregate & become cross-linked to form collagen fibers
- ✓ Collagen synthesis starts with activation genes for α -peptide chains in DNA
- ✓ There are multiple genes to synthesize d/t types of α -peptide chains

Cont...

➤ Steps of Collagen Biosynthesis

- Collagen is one of many proteins that normally function outside of cells
- Collagen biosynthesis involves **two steps**:
 1. Intracellular steps
 2. Extracellular steps

Cont...

1. Intracellular steps

step1:- Formation of pro- α chains

- ✓ Newly synthesized polypeptide precursors of α - chains (prepro- α chains)
 - Contain special amino acid sequence at their N-terminal ends
- ✓ N-terminal amino acid sequence:
 - Rapidly cleaved in RER to yield a precursor of collagen called **pro- α chain**

step2: Hydroxylation:

- ✓ Within lumen of RER Pro- α chains are processed by number of enzymatic steps **Pro** & **Lys** residues found in **Y-position** of **-Gly-X-Y-** sequence undergo hydroxylation by Vitamin C dependent enzymes (Prolyl hydroxylase & Lysyl hydroxylase)

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Step3:- Glycosylation:

- ✓ Some Hydroxylysine residues are modified by enzymatic glycosylation with **glucose** or **glucosyl-galactose**

Step4:- Assembly & secretion

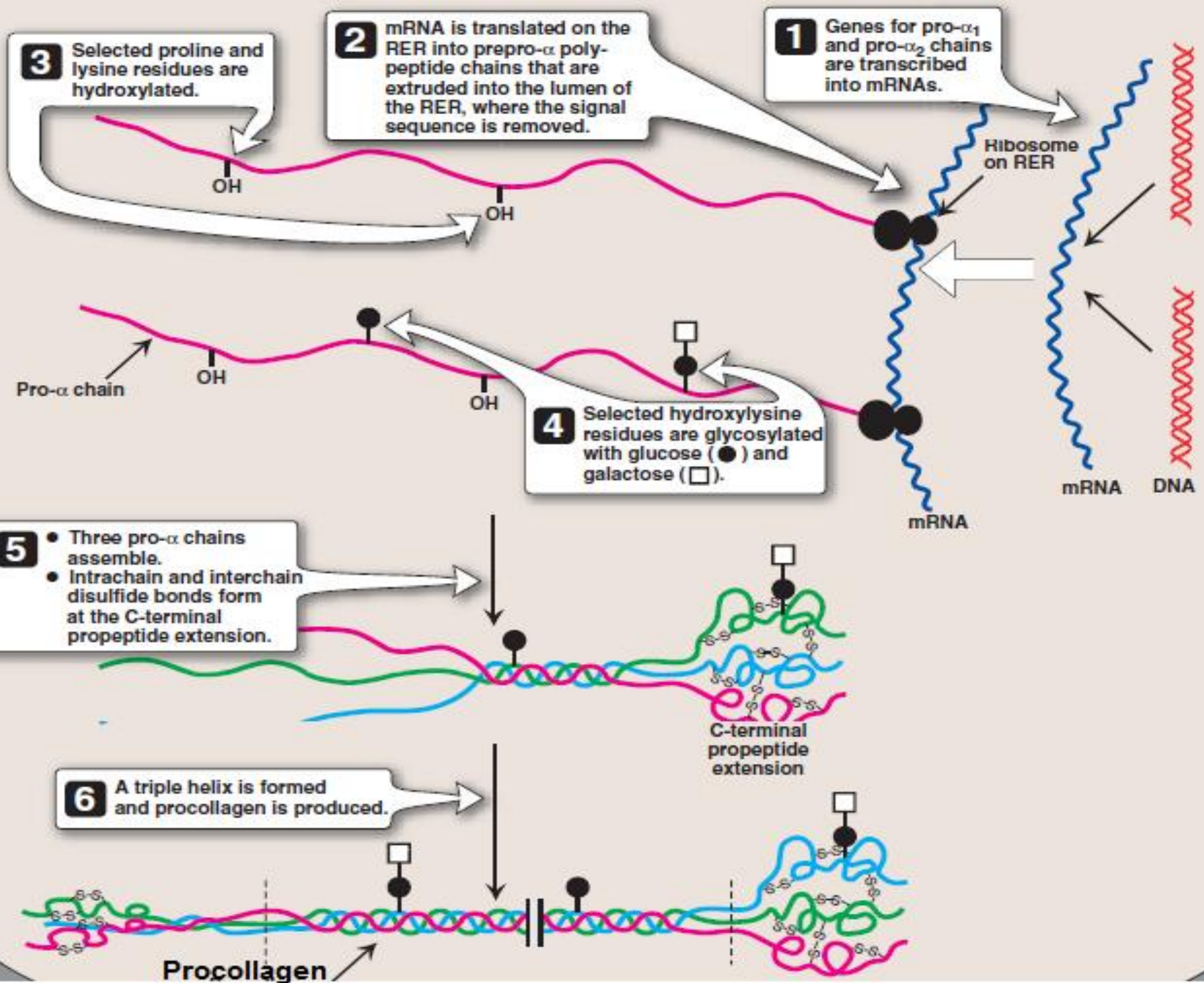
- ✓ **Assembly:**

- After hydroxylation & glycosylation:

- ❖ **Interchain disulfide bonds** formed b/n three pro- α chains at C-terminal & bring three pro- α chains into alignment favorable for **helix formation**

- ❖ **Procollagen** (Precursor of collagen that has central region of triple helix flanked by non-helical N- & C-terminal extensions) is formed

- ❖ Non-helical N- & C-terminal extensions of procollagen are called **propeptides**



Cont...

✓ Secretion:

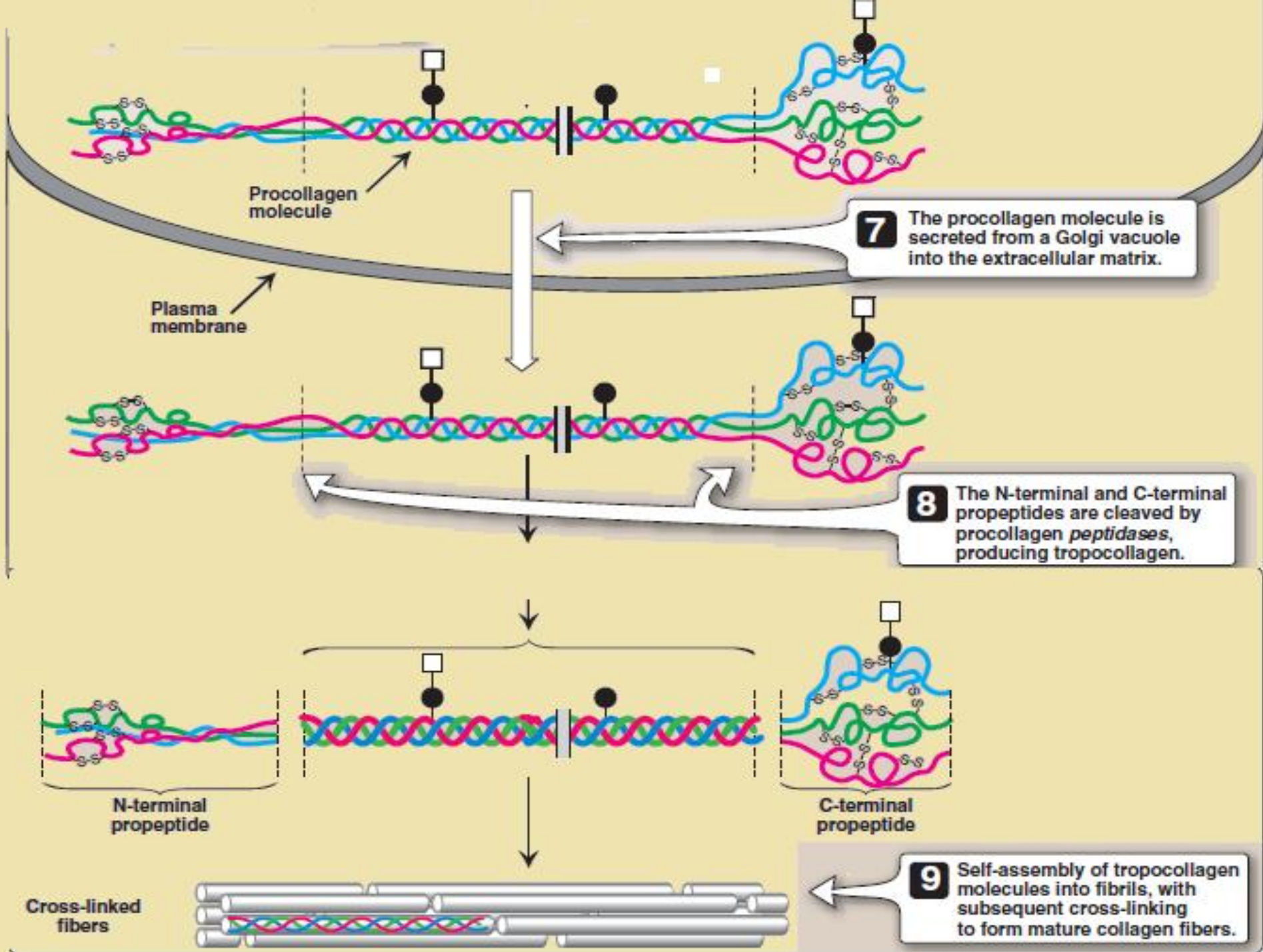
- ❖ Procollagen moves through **Golgi apparatus** → cell membrane, causing release of procollagen to extracellular space

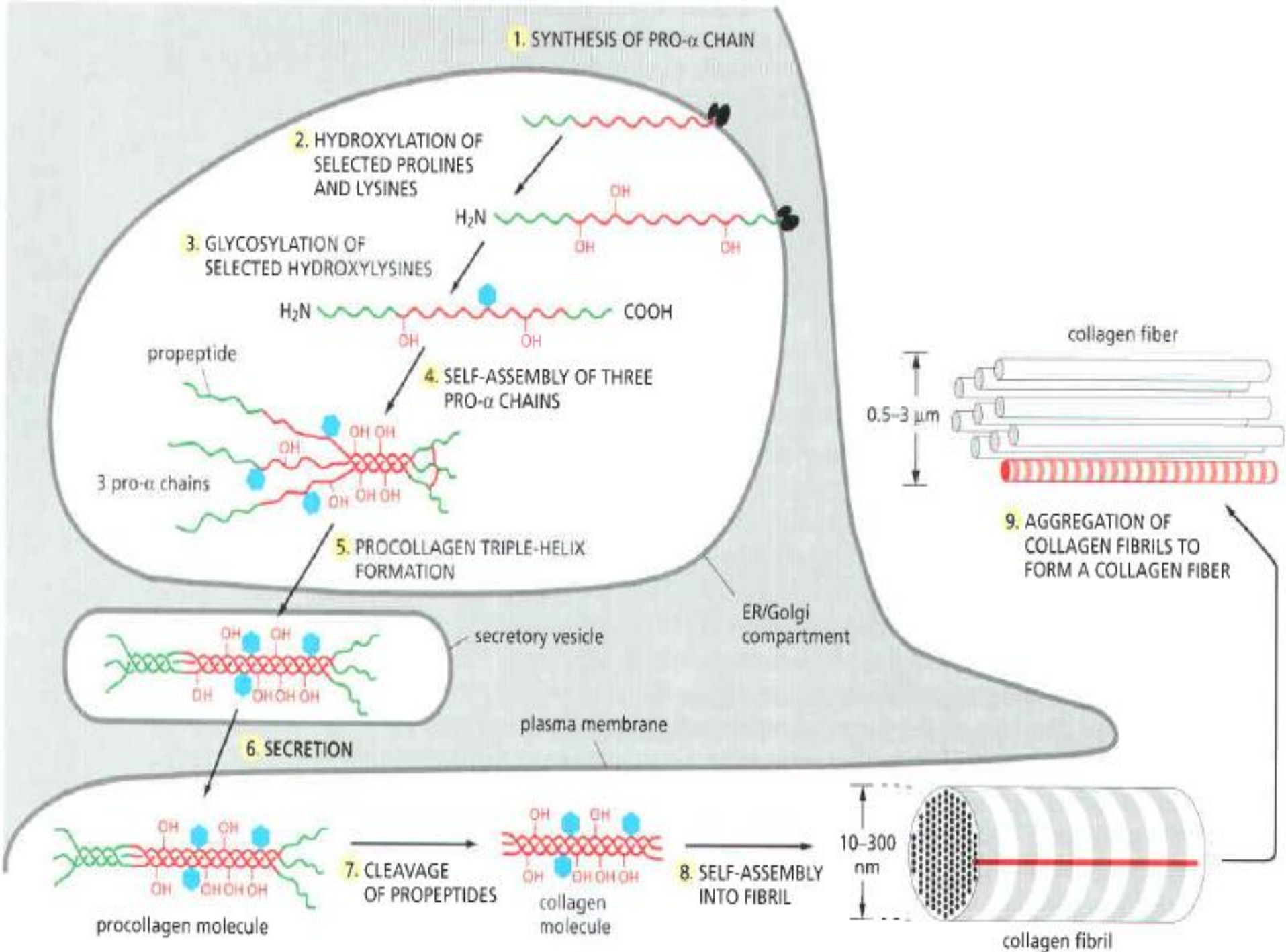
2. Extracellular Steps

I. Extracellular cleavage of procollagen: N- & C-procollagen peptidases remove terminal propeptides, releasing triple-helical molecule → **Tropocollagen**

II. Formation of collagen fibrils: by each tropocollagen molecules overlapping its neighbor

III. Cross-link formation: to form → Mature collagen fibers





Osteogenesis Imperfecta

- ✓ Gly founds in every third position of α -polypeptide chain
- ✓ Gly due to its simple R-group facilitates proper twisting of three α -polypeptide chains during triple helix formation
- ✓ Mutations in gene for Gly residues in pro- α 1 or pro- α 2 chains of type I collagen cause replacement of Gly residues (-Gly-X-Y-) by amino acids with bulky side chains
- ✓ Resultant structurally abnormal pro- α chains prevent formation of required triple-helical conformation
- ✓ Final develop **Osteogenesis Imperfecta** (Brittle Bone syndrome)
- ✓ This disorders distinguished by bones that easily bend & fracture

Osteogenesis Imperfecta

Type-I: Osteogenesis imperfecta tarda

- ✓ Due to moderate mutation leading to decreased production of $\alpha 1$ & $\alpha 2$ chains of type-I collagen
- ✓ Moderate form Osteogenesis imperfecta in w/c baby will be born (defect is compatible with life)
- ✓ Secondary to minor trauma, infant gets multiple bone fractures
- ✓ Patients may have **blue sclera** due to tissue thinning that allows underlying pigment to show through

Type-II: Osteogenesis imperfecta Congenita

- ✓ Severe form of osteogenesis imperfecta in w/c product of conception suffer from multiple mutation, Intra-utrine death

»Scurvy:



- In **ascorbic acid(Vit. C)** deficiency hydroxylating enzymes, prolyl hydroxylase & lysyl hydroxylase, are unable to function
- Lack of prolyl & lysyl hydroxylation
 - Inter chain H-bond formation is impaired → Formation of unstable triple helix
 - Collagen fibrils cannot be cross-linked, greatly decreasing tensile strength of assembled fiber
- Patients often show bruises on limbs as result of subcutaneous extravasation of blood due to capillary fragility

»Scurvy:

➤ Deficiency of ascorbic acid

– Results:

- Disease characterized by sore & spongy gums, loose teeth, fraaile blood vessels, swollen ioints, & anemia



Cont...

» Ehlers-Danlos syndrome (EDS)

- Heterogeneous group of disease in w/c extracellular processing of collagen is disturbed
- Causes:
 - Inherited deficiencies of extracellular collagen-processing enzymes
- Defects in collagen type I fibrils processing
→ Fragile, **stretchy skin** & loose joints



Thank U!